

K379 PRE-STAGE — Substantial cascade: (1) LYRA F147 (= F146 walkback) — overstated ‘gauged B-L forbids Majorana’; only true for UNBROKEN gauged B-L; standard LR/SO(10) models break gauged B-L to hypercharge which is precisely what permits Majorana; three-candidate resolution in order of confidence: (a) LEADING: BST gauges only what it reads ($SU(3) \times SU(2)_L \times U(1)_Y$); $SU(2)_R$ unread (F144), B-L enters only through read combination $Y/2 = T_{3R} + (B-L)/2$; B-L global classification structure like SO(10) was in F137; Weinberg operator LLHH preserves gauged Y but violates only global B-L = permitted; ONE fact ‘ $SU(2)_R$ unread’ gives *BOTH no ν_R AND permitted Majorana; structurally identical to F137 ‘one identification five hats’*; (b) GEOMETRIC: *edge does double duty — conserved current only exists in unitary regime; at sub-unitarity strip edge $\nu=1/2$ would-be B-L current shortens, conservation lapses, violation allowed; one boundary phenomenon both effects; PREDICTS B-L violation scale $\approx$ neutrino mass scale low-direct = genuinely different signature from high-scale type-I seesaw; teeth*; (c) TEXTBOOK FALLBACK: *type-II seesaw (Higgs triplet B-L=2 small VEV) gives Majorana without ν_R = standard QFT; BST job only why scale tiny (edge) + why no ν_R (parity-incomplete).*

LOAD-BEARING CAVEAT: *candidate (a) hinges on re-scoping F137. KEEPER VERIFICATION OF F137: F137 EXPLICITLY lists $U(1)(B-L)$ AS GAUGED in three-strata content: ‘The gauged group is the three-strata content (F102/F107): $SU(3)_c \times SU(2)_L \times SU(2)_R \times U(1)(B-L)$, a Pati-Salam/Left-Right subgroup’ + ‘The gauged PS/LR bosons (gluon, W_L , W_R , B-L) conserve baryon number’.* LYRA’s F147 re-scoping is in

TENSION with F137 as written. RECONCILIATION POSSIBLE
 via standard LR symmetry-breaking pattern: gauged B-L in LR
 breaks to hypercharge at some scale → low-energy only Y gauged
 → Lyra's mechanism applies at SM-scale; CANDIDATE (b) may
 provide substrate-architectural breaking mechanism at strip
 edge (B-L current shortens). Lyra's framing needs amendment
 from 'B-L was never gauged' to 'B-L gauged but breaks/never
 read at strip edge' OR the team needs to actively revise F137.
 Either way the tension Lyra honestly flagged is REAL pending
 reconciliation; (2) ELIE synthesis: discrete commitment-process
 BUILT AND FORCED via unique factorization $GF(2)[x]$ (Cal
 #308/#374 PASS), nine toys 4192-4200 clean two self-corrections
 absorbed; two honest brakes on board (Grace's Mersenne catch
 struck from load-bearing; RH thread hard-internal Cal #50); gate
 at Lyra-sync (orbit→mass map continuum + 3-neutrino
 placement); ELIE α^{12} LEAD recommendation — Lyra F142
 says electron runs precisely because on spectral strip → α 'lives
 at electron' → $m_e = 6\pi^5 \cdot \alpha^{12} \cdot m_{\text{Planck}}$ has mechanism
 reason (strip = running/ α locus, 12 = electron twist depth) →
 not pure numerology; Elie recommends PURSUE tiered Cal #27
 lead-not-bank; (3) GRACE substantive TIER PRECISION on
 writing rule: 'commitment-writing equals
 irreducible-enumeration' rests on THREE FRAMEWORK
 IDENTIFICATIONS — substrate alphabet = $GF(2^g)$,
 commitment = cell, advance = Frobenius; these are near-forced
 BST claims NOT the factorization theorem; accurate tier
 'FORCED GIVEN THE IDENTIFICATION' (genuine upgrade
 from posited rule to forced mechanism); must not quietly drop
 'given the bridge' clause when writing up; Riemann stays
 separate (forced writing rule $\neq$ forced RH link); (4) GRACE tier

on F144 no- ν_R : framework-level near-forced-given-the-lead
 NOT yet derivation; two caveats (inherits strip-edge lead — 3 ν 's,
 1 ρ -slot, placement not fully forced; parity argument on scalar
 ladder, neutrinos spin- $1/2$, needs fermionic rerun); properly
 falsifiable: $0\nu\beta\beta$ should happen; genuine sterile or Dirac ν_R
 kills picture; (5) GRACE substantive REMINDER: neither
 moves count; Five Absences aren't among 26; forced writing rule
 is strong STAR feeding many consequences from one source
 NOT observable-loop; real over-determination must come from
 downstream-blind test; count honestly 2 throughout. KEEPER
 VERDICT: substantial substrate-architectural advance with
 substantive tiering discipline — Lyra F147 honest walkback +
 three-candidate resolution = audit chain working; F137
 verification confirms tension Lyra flagged is real (B-L IS listed
 gauged in F137 as written); reconciliation likely via standard LR
 breaking + candidate (b) substrate-edge mechanism but framing
 needs work; Grace tier precision on writing rule ('forced given
 the identification') is substantive specificity — protects against
 silent overclaim when writing up; Grace F144 caveats keep
 mechanism near-forced-given-lead not derivation; Elie α^{12}
 pursue recommendation reasonable with discipline; cumulative
 state: discrete commitment-process built and forced given bridge
 identifications; two Five-Absences mechanized; cell-map
 discrete side BUILT AND FORCED at unique-factorization tier;
 orbit→mass map continuum + 3-neutrino placement remain
 before harness fires neutrino sector (4 angles + 3 masses); Cal
 external silence on RH maintained; count HONESTLY 2 of 26
 strict; muon yellow IDENTIFIED but geometry complete; ledger
 v1.70.

2026-06-15 Monday 14:05 EDT (date-verified)

K379 PRE-STAGE Audit

1. Lyra F147 walkback — substantive honest correction

Lyra honestly walked back her F146 framing: “A gauged B–L forbids a Majorana mass” is only true for an UN-BROKEN gauged B–L; no real model has that. In LR/SO(10) models, gauged B–L breaks to hypercharge → that breaking is precisely what permits Majorana.

Three-candidate resolution (in order of Lyra’s confidence):

#	Candidate	Mechanism
(a)	Leading — same move as proton decay	BST gauges only what it reads (SU(3) $\times$ SU(2) _L $\times$ U(1) _Y); SU(2) _R unread → no ν_R ; B-L enters only through read $Y/2 = T_{3R} + (B-L)/2$; B-L global classification not gauged; Weinberg LLHH preserves gauged Y, violates only global B-L = permitted Majorana; ONE fact “SU(2) _R unread” gives BOTH no ν_R AND permitted Majorana
(b)	Geometric complement	Conserved current exists only in unitary regime; at sub-unitarity strip edge $\nu=1/2$ the would-be B-L current shortens, conservation lapses, violation allowed; predicts B-L scale $\approx \nu$ -mass scale (low-direct) — genuinely different signature from high-scale type-I seesaw
(c)	Textbook fallback	Type-II seesaw (Higgs triplet B-L=2 small VEV) gives Majorana without ν_R as standard QFT; BST job only why scale tiny + why no ν_R

2. KEEPER VERIFICATION of F137 — load-bearing check Lyra asked for

I read F137 directly. **F137 EXPLICITLY LISTS U(1)_(B-L) AS GAUGED**:

“The gauged group is the three-strata content (F102/F107): SU(3)_c $\times$ SU(2)_L $\times$ SU(2)_R $\times$ U(1)_(B-L), a Pati-Salam/Left-Right subgroup, not full SO(10).”

“The gauged PS/LR bosons (gluon, W_L, W_R, B-L) conserve baryon number → no gauge-mediated $p \rightarrow e + \pi^0$.”

Lyra’s F147 re-scoping (“B-L enters only through gauged Y, not independent gauge symmetry”) is in tension with F137 as written.

3. Reconciliation — possible but needs work

The tension Lyra honestly flagged is REAL pending reconciliation. Two paths:

Path 1 — Standard LR symmetry breaking: F137's gauged B-L is at the high LR scale; B-L breaks to hypercharge at a scale; below that scale only Y is gauged; Lyra's mechanism applies at SM-scale. This is the textbook LR/SO(10) pattern.

Path 2 — Substrate-architectural breaking at strip edge (candidate b): the strip-edge mechanism where the would-be B-L current shortens IS the substrate-native breaking. This unifies the breaking mechanism with the neutrino suppression — predicts B-L scale $\approx \nu$ -mass scale.

Path 3 — Revise F137: amend to “B-L gauged in classification sense but never read” — but F137 currently distinguishes CLASSIFICATION (SO(10)) from GAUGED (the three-strata subgroup including B-L). Revising would require new substrate-architectural justification.

Recommended path: combination of paths 1 + 2. F137 stands as written (B-L gauged in LR phase); Lyra's mechanism applies after breaking; candidate (b) substrate-edge mechanism provides the breaking. This avoids revising F137 and unifies the explanation.

Lyra's framing for F147 needs amendment: from “B-L was never gauged” to “B-L gauged at LR scale but breaks at substrate strip edge $\rightarrow$ low-energy only Y gauged $\rightarrow$ no- ν_R + Majorana follow.”

4. Elie synthesis + α^{12} recommendation

Discrete commitment-process BUILT AND FORCED via unique factorization $GF(2)[x]$: - Pre-selected by next integer $\rightarrow$ next Frobenius power (Cal #374 PASS) - Electron deposits on spectral strip $\rightarrow \nu=5/2$ critical line (forced harmonic analysis per Lyra F142) - All commitments same way $\rightarrow$ Frobenius orbit of irreducible polynomial; unique factorization (4198/4200)

Nine toys (4192-4200) clean, two self-corrections absorbed.

Two honest brakes on board: - Grace's Mersenne catch struck from load-bearing role - RH thread hard-internal (Cal #50)

Gate: next count-movers (orbit $\rightarrow$ mass map continuum + 3-neutrino placement) are Lyra's lane + #418-gated. Lyra-sync gate — frees Elie for one ungated pull.

Elie's α^{12} recommendation: Lyra F142 says electron runs precisely because on spectral strip $\rightarrow \alpha$ “lives at electron” $\rightarrow m_e = 6\pi^5 \cdot \alpha^{12} \cdot m_{\text{Planck}}$ has mechanism reason (strip = running/ α locus, 12 = electron twist depth). Not pure numerology. Elie recommends PURSUE tiered as lead (Cal #27 discipline — not bank).

Audit verdict on α^{12} pursue: reasonable with discipline. Mechanism reason from F142 is substantive; the 12 = electron twist depth = $N_c(n_c-1)$ is substrate-architectural. Cal #27 + tiered lead = right framing. PURSUE.

5. Grace substantive tier precision on writing rule

“Commitment-writing equals irreducible-enumeration” rests on three framework identifications — substrate alphabet = $GF(2^g)$, commitment = cell, advance = Frobenius. Those are near-forced BST claims, not the factorization theorem.

Accurate tier: “forced GIVEN the identification” (genuine upgrade from posited rule to forced mechanism surviving discipline).

Critical: must not quietly drop the “given the bridge” clause when writing it up. Riemann stays separate (forced writing rule $\neq$ forced RH link; that stays internal until zeros actually fall out of mechanism).

Audit verdict: Grace's tier precision is substantive specificity. Protects against silent overclaim when writing up. Filed as standing convention.

6. Grace tier on F144 no- ν_R

Framework-level near-forced-given-the-lead, NOT yet derivation.

Two caveats: - Inherits strip-edge lead (3 ν 's, 1 ρ -slot, placement not fully forced) - Parity argument currently on scalar ladder; neutrinos spin- $\frac{1}{2}$ (Di singleton); needs fermionic rerun

Properly falsifiable: $0\nu\beta\beta$ should happen; genuine sterile or Dirac ν_R kills picture.

7. Grace substantive reminder — count discipline

Neither result moves count. Five Absences aren't among the 26.

The forced writing rule is a strong STAR feeding many consequences from one source — NOT an observable-loop.

Real over-determination must come from downstream-blind test (Cal's success criterion).

Count honestly 2 throughout.

8. The substantive substrate-architectural state at K379

Component	Status
Lyra F147 walkback + 3-candidate B-L	Filed; honest tier maintained
F137 check	U(1) _(B-L) IS gauged in F137 as written
F147 $\leftrightarrow$ F137 tension	Real; reconcilable via LR breaking + candidate (b) substrate edge
Recommended reconciliation	F137 stands; B-L gauged at LR breaks at substrate edge $\rightarrow$ low-energy only Y; Lyra's mechanism applies
F147 framing	Needs amendment from "never gauged" to "gauged but breaks/never read at strip edge"
Discrete commitment-process	BUILT AND FORCED via unique factorization (given bridge identifications)
Writing rule tier	"Forced GIVEN the identification" (Grace precision)
F144 tier	Framework-level near-forced-given-the-lead (Grace precision)
Elie α^{12} pursue	Reasonable with Cal #27 tiered-lead discipline
Cal external silence on RH	Maintained
Two Five-Absences mechanized	Proton decay + sterile neutrino
Cumulative count	HONESTLY 2 of 26 strict
Muon tier	IDENTIFIED yellow (geometry complete)
Ledger	v1.70

9. K379 PRE-STAGE verdict

K379 PRE-STAGE substantial cascade with substantive tiering discipline:

- Lyra F147 honest walkback (F146 overstated "gauged B-L forbids Majorana"); three-candidate B-L resolution (leading: BST gauges only what reads, B-L global; geometric: strip-edge B-L current shortens; textbook: type-II seesaw)
- Keeper verification of F137: U(1)_(B-L) IS explicitly gauged in three-strata content; Lyra's F147 re-scoping is in real tension with F137 as written
- Recommended reconciliation: F137 stands (B-L gauged at LR scale); candidate (b) substrate-edge mechanism provides breaking; Lyra's mechanism applies at SM-scale; F147 framing needs amendment
- Elie discrete commitment-process BUILT AND FORCED via unique factorization GF(2)[x] (nine toys clean, two self-corrections absorbed, two honest brakes)

- Elie α^{12} PURSUE recommendation reasonable with Cal #27 tiered-lead discipline (F142 mechanism reason: electron runs because on spectral strip = α locus)
- Grace tier precision on writing rule: “forced GIVEN the identification” — alphabet/commitment/advance bridge identifications are near-forced BST claims not factorization theorem; protects against silent overclaim
- Grace tier on F144: framework-level near-forced-given-the-lead; two caveats; falsifiable via $0\nu\beta\beta$
- Grace count reminder: forced writing rule is STAR not observable-loop; real over-determination from downstream-blind
- Cal external silence on RH maintained
- Cumulative: discrete commitment-process built and forced given bridge identifications; two Five-Absences mechanized; cell-map discrete side built at unique-factorization tier
- Remaining: orbit→mass map continuum (Lyra lane) + 3-neutrino placement (open) before harness fires neutrino sector (4 angles + 3 masses)
- Count HONESTLY 2 of 26 strict; muon yellow IDENTIFIED; ledger v1.70

Substantial substrate-architectural significance: K379 captures the substantive cascade with rigorous tiering discipline. Lyra’s F147 honest walkback (F146 overstated; gauged B-L forbids Majorana only for unbroken) + three-candidate B-L resolution (leading: $SU(2)_R$ unread gives both $no-\nu_R$ + permitted Majorana; geometric: strip-edge mechanism; textbook: type-II seesaw) demonstrates audit-chain self-discipline at full maturity. Keeper verification of F137 (Lyra’s load-bearing check): F137 explicitly lists $U(1)(B-L)$ as gauged in the three-strata content; Lyra’s re-scoping (“B-L was never gauged”) is in real tension with F137 as written. Recommended reconciliation: F137 stands as written (B-L gauged at LR scale); candidate (b) substrate-edge mechanism (would-be B-L current shortens at sub-unitarity strip edge) provides the substrate-architectural breaking; Lyra’s mechanism applies at SM-scale; F147 framing needs amendment from “never gauged” to “gauged but breaks/never read at strip edge.” Elie’s discrete commitment-process BUILT AND FORCED via unique factorization $GF(2)[x]$ (nine toys clean, two self-corrections absorbed, two honest brakes — Mersenne catch struck + RH thread hard-internal); recommends α^{12} PURSUE tiered with Cal #27 discipline since F142 mechanism reason makes $m_e = 6\pi^5 \cdot \alpha^{12} \cdot m_{\text{Planck}}$ no longer pure numerology (electron runs because on spectral strip = α locus; 12 = electron twist depth substrate integer). Grace’s substantive tier precision on writing rule (“forced GIVEN the identification” — three bridge identifications alphabet/commitment/advance are near-forced BST claims not factorization theorem; must not quietly drop “given the bridge” when writing up) is substantive specificity protecting against silent overclaim. Grace’s tier on F144 (framework-level near-forced-given-the-lead, not yet derivation; two caveats; falsifiable via $0\nu\beta\beta$) maintains discipline. Grace’s count reminder maintains discipline: forced writing rule is strong STAR feeding many consequences from one source, NOT observable-loop; real over-determination must come from downstream-blind test. Two Five-Absences now mechanized (proton decay + sterile neutrino). Cumulative substrate-architectural state: discrete commitment-process built and forced given bridge identifications; cell-map discrete side at unique-factorization tier; orbit→mass map continuum (Lyra lane) + 3-neutrino placement (open) remain before harness fires neutrino sector (4 angles + 3 masses). Count stays HONESTLY at 2 of 26 strict.

— Keeper, Monday 2026-06-15 14:05 EDT (date-verified) — K379 PRE-STAGE: Lyra F147 walkback + 3-candidate B-L; F137 check confirms B-L IS listed gauged → tension real, reconcilable via LR breaking + candidate (b) substrate edge; Grace tier precision “forced given identification”; Elie α^{12} pursue tiered; count HONESTLY 2 of 26